

Wind Field Generation

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Lecture #7
Controller Design for Wind Turbines and Wind Farms

Hochschule Flensburg
University of Applied Sciences

SWE Stuttgart Wind Energy @ Institute of Aircraft Design

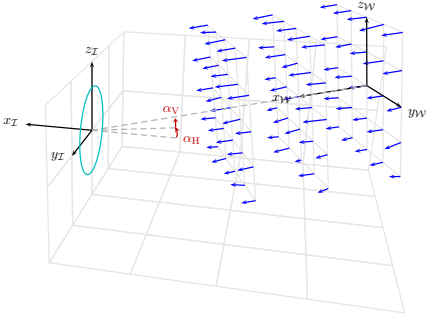
Wind Energy Technology Institute

Schedule

28.09.2026	1	Controller Design Objectives and Modeling
28.09.2026	2	Baseline Generator Torque Controller
12.10.2026	3	Collective Pitch Controller
19.10.2026	4	Filter Design
26.10.2026	5	Tower Damper
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09.11.2026	7	Wind Field Generation
16.11.2026	10	Lidar-Assisted Control I
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30.11.2026	8	Steady State Calculations
07.12.2026	9	Individual Pitch Control
14.12.2026	12	Wind Farm Effects



Wind Field Generation



Motivation

- Wind is not totally random.
- Understanding of wind is crucial in wind energy!

Main questions

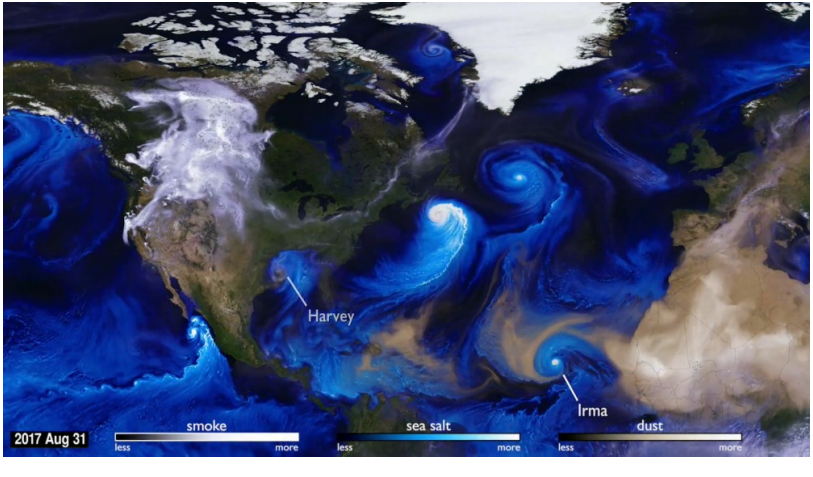
- How is a turbulent wind field generated?
- How can we generate rotor-effective wind speed?

Contents

1. Origin of Wind
2. Atmospheric Boundary Layer
3. Wind Field Generation
4. Research Challenges in Wind Energy
5. Conclusion and Learning Objectives



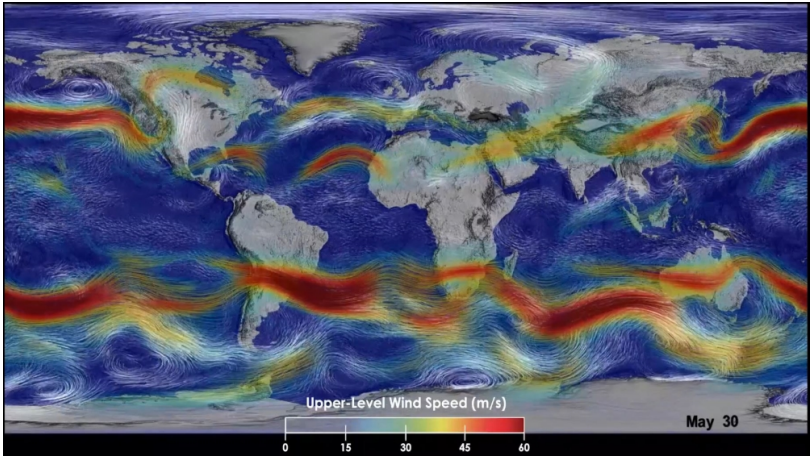
Dust in the wind



[NASA Visualization Explorer]

- Saharan dust is swept into the outer cyclonic winds of tropical storm Irma.
- Rain at the center of the storm washes it out of the air.

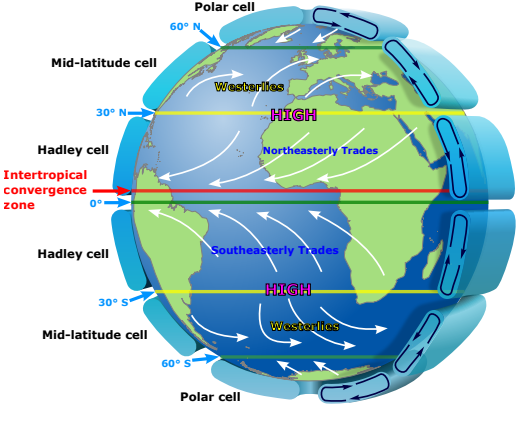
Global wind



[Surface and Upper-Level Winds]

- High altitude winds circumnavigate the earth in wave-like paths from west to east.
- Winds closer to the ground are trade winds, west winds and polar east winds.

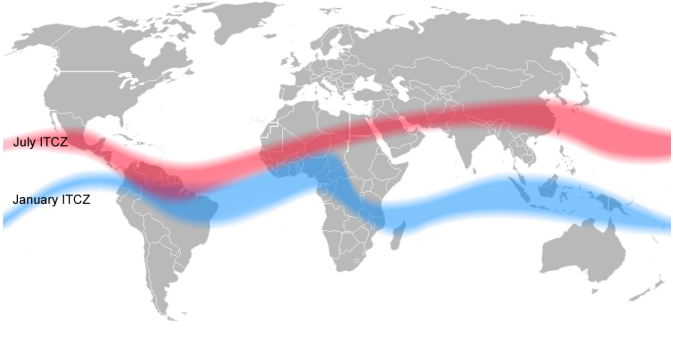
Atmospheric circulation



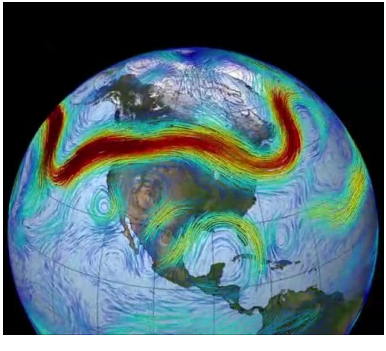
based on [Wikipedia]

- As a result of different levels of solar heating, warm air rises at the equator and falls at the poles.
- Wind as a compensatory movement
- Because of the rotation of the earth (Coriolis force), movements are deflected to the right in the northern hemisphere and to the left in the southern hemisphere.
- Three wind systems through surface convergence

Inter-tropical convergence zone



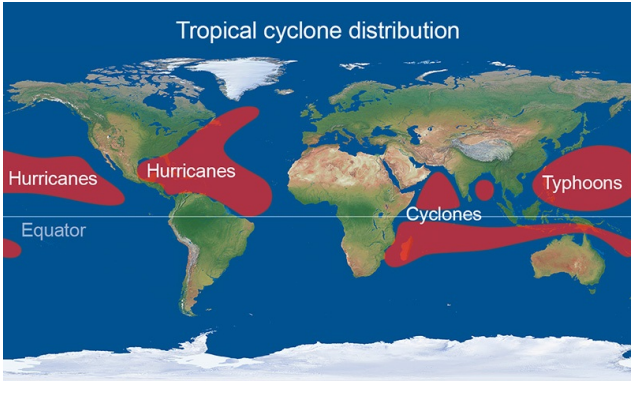
[Wikipedia]



[NASA/Goddard Space Flight Center]

- Also equatorial low pressure belt or ITCZ (Inter-tropical Convergence Zone)
- Due to different land mass distribution and solar warming it is not constantly at 0°.
- Trade winds on both sides of the ITCZ for rapid ocean crossing
- A hurricane, cyclone or typhoon often occurs here.
- Westerly winds to compensate: polar front jet stream (60°), subtropical jet stream (30°)

Location and naming of tropical cyclones



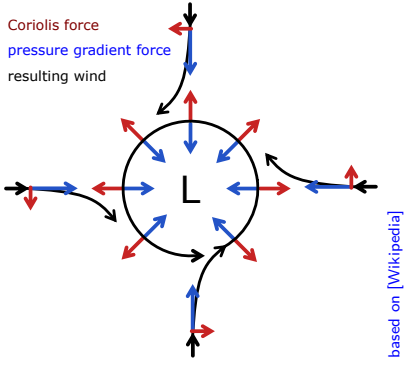
[www.metoffice.gov.uk]

- Hurricanes - Atlantic and North-East Pacific Oceans
- Typhoons - North-West Pacific Ocean
- Cyclones - South-West Pacific and Indian Ocean

Geostrophic wind



[NASA visible earth]

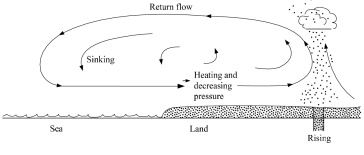


based on [Wikipedia]

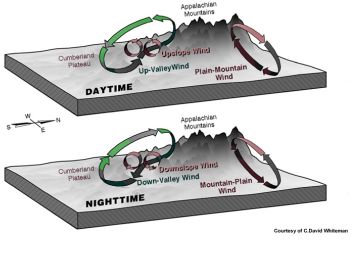
- Air flow at high altitudes unaffected by ground friction
- Balance between gradient force and Coriolis force
- Currents around low pressure areas are counterclockwise in the northern hemisphere.

Video: Coriolis force

Local winds



[1]



[Wikipedia]

Sea-land circulation

- Daily cycle due to different heat capacity
- Sea wind during the day: land warms up faster
- Land wind at night: land cools down faster

Mountain-valley circulation

- Daily cycle due to different proximity to the ground
- Valley wind during the day: the air on the mountain slope warms up
- Mountain wind at night: air on the mountain slope cools down

Formation of the atmospheric boundary layer

Atmospheric boundary layer

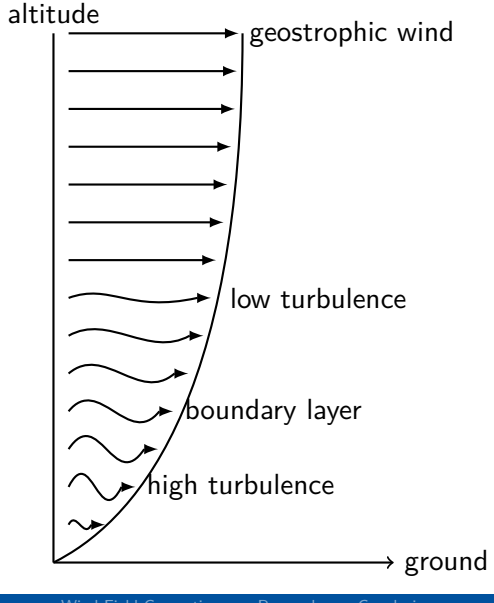
- High altitude $\rightarrow$ geostrophic wind
- Ground $\rightarrow$ no wind speed

Energy extraction by ground roughness

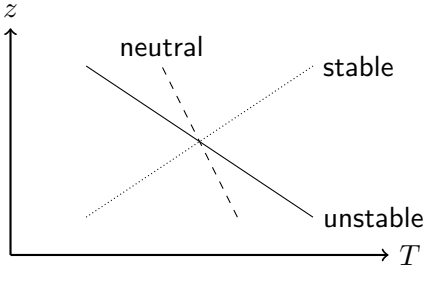
- Vertical wind speed gradient
- Turbulent momentum and mass exchange with higher air layers

Boundary layer thickness

- From $100\,\mathrm{m}$ on cold nights
- Up to $2000\,\mathrm{m}$ on warm summer days with low wind speed
- On average approx. $1000\,\mathrm{m}$



Thermal stratification and atmospheric stability



Unstable

- Air near the ground is warmer than the air above.

→ There is more vertical mass transfer and thus increased turbulence.

Stable

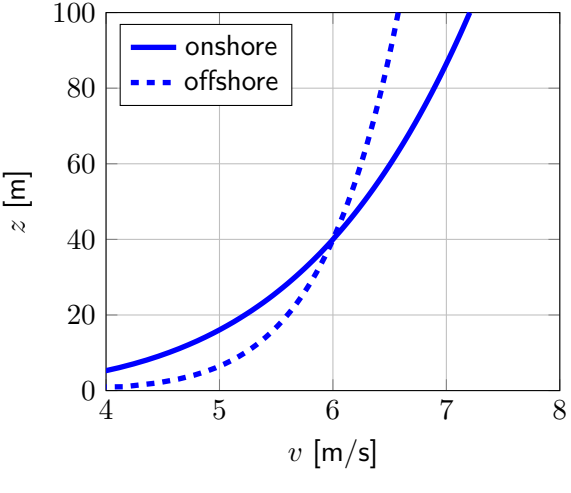
- Air near the ground is colder than the air above.

→ There is less vertical mass transfer and turbulence is suppressed.

Neutral

- Adiabatic temperature profile (no heat interaction, only caused by pressure)
- Air temperature decreases by about 1°C per 100 m increase in altitude.
- Often occurs at high wind speeds

Exponential wind profile



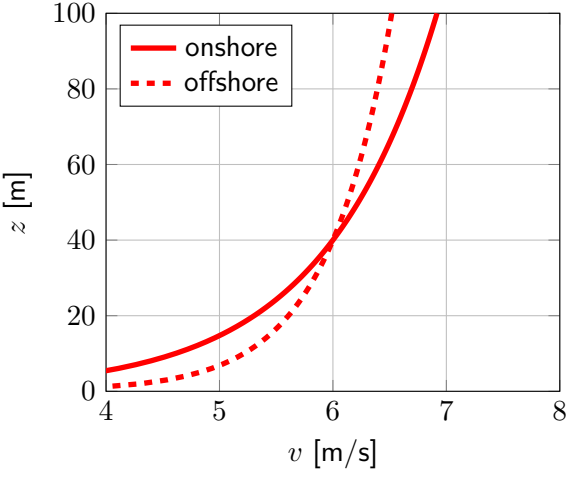
$$v_2 = v_1 \left(\frac{z_2}{z_1}\right)^{\alpha}$$

v_1 wind speed at height z_1
v_2 wind speed at height z_2
α roughness exponent

Roughness exponent α:

0.1	desert, open sea
0.15	meadows, pastures
0.2	little vegetation
0.37	cities, forest

Logarithmic wind profile



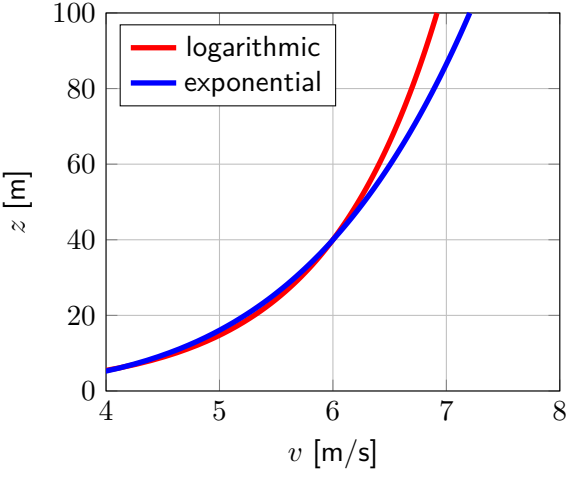
$$v_2 = v_1 \frac{\ln(z_2/z_0)}{\ln(z_1/z_0)}$$

v_1 wind speed at height z_1
v_2 wind speed at height z_2
z_0 roughness length

Roughness length z_0

0.001 m	desert, open sea
0.01 m	meadows, pasture
0.1 m	little vegetation
1 m	cities, forest

Logarithmic and exponential wind profile



$z = 40\,\text{m}$ and $v = 6\,\text{m/s}$

$z_0 = 0.1\,\text{m}$
$\alpha = 0.2$

- Logarithmic: physical, often for measurement data
- Exponential: empirical, often for simulations

Influence of thermal stratification on wind profile

Neutral thermal stratification

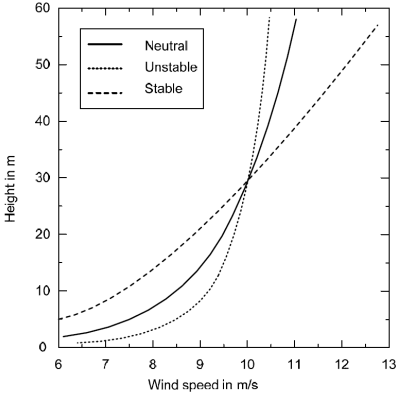
Pure logarithmic wind profile.

Stable thermal stratification

Low vertical momentum and mass exchange. $\rightarrow$ Greater gradient of wind speed with increasing height.

Unstable thermal stratification

Rising air leads to strong vertical momentum and mass exchange. $\rightarrow$ Smaller gradient of wind speed with increasing height.



[1], [Video DTU]

Turbulence

Terminology:

- Turbulence describes a flow condition of the air that is characterized by disorder.
- Turbulence intensity describes the ratio of the standard deviation of the wind speed to the mean value of the wind speed related to a time interval of 10 minutes.



[Leonardo da Vinci], ca. 1510-12



3D illustration by [Paolo Colagrossi]

The Starry Night

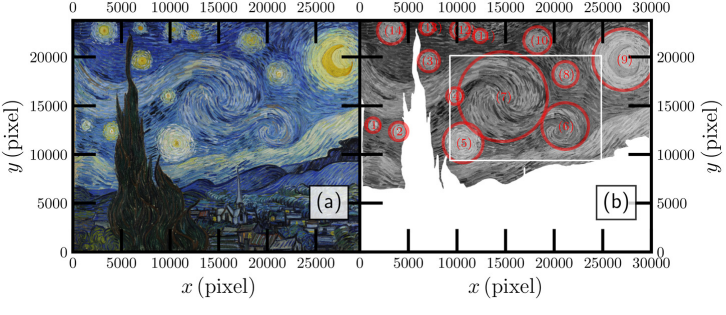
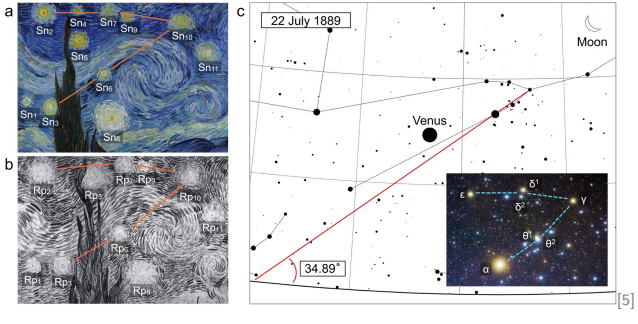


FIG. 2. (a) A high-resolution van Gogh's ***The Starry Night*** obtained from https://artsandculture.google.com with a size of 92.1 cm × 73.7 cm and 30 000 pixel ×23 756 pixel. Visually, the sky seems to be flowing with swirling eddies. (b) Gray version of the ***The Starry Night***, where the region studied by Finlay[16] is illustrated by a white square. The non-flow part is masked out manually. The whirls/eddies are recognized by naked eyes.

[2]

- Famous painting by Vincent van Gogh during his time in the asylum 1889
- Van Gogh's "Starry Night" might align with physics, see [2], TEDEd, and nature.com.
- Early 2025: conclusions in [2] might be completely unfounded, see [3] and phys.org.
- Late 2025: [2] wrongly treated paint strokes as actual fluids and cherry-picked the image data to force a mathematical match [4].

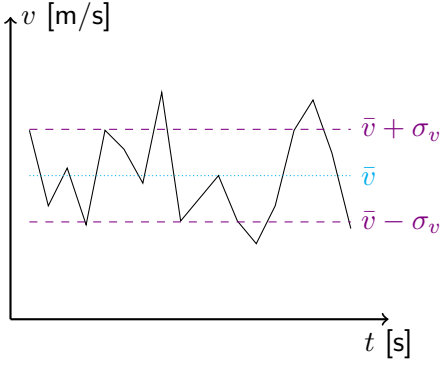
The Starry Night 2026



[5]

- The study [5] models the painting's sky with astronomical and archival data.
- Celestial alignment dates the sky to dawn on ∼22 July 1889 rather than June.
- The sky was an observed reality, giving art history a reproducible method.

Turbulence intensity



Mean wind speed $\bar{v}$

$$\bar{v} = \frac{1}{n} \sum_{i=1}^{n} v_i$$

Variance σ_v^2 and standard deviation σ_v

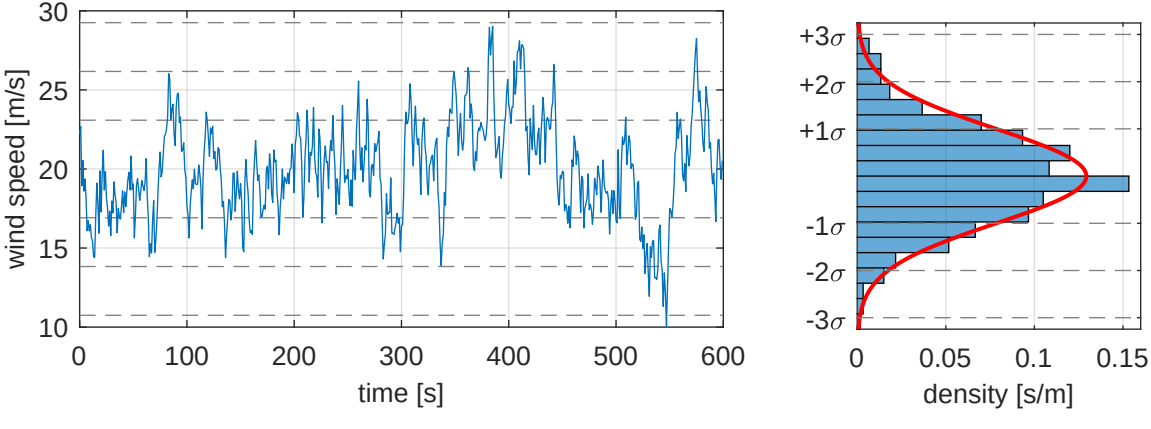
$$\sigma_v^2 = \frac{1}{n-1} \sum_{i=1}^{n} (v_i - \bar{v})^2 \qquad \text{Bessel's corr.}$$

$$\sigma_v = \sqrt{\sigma_v^2}$$

Turbulence intensity

$$\mathrm{TI} = \frac{\sigma_v}{\bar{v}}$$

Wind is considered normal distributed acc. to IEC 61400-1



- The turbulent velocity fluctuations are assumed to be a stationary, random vector field whose components have zero mean Gaussian statistics [6].
- Empirical rule: 68%, 95%, 99.7% of the data are within $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$, respectively.

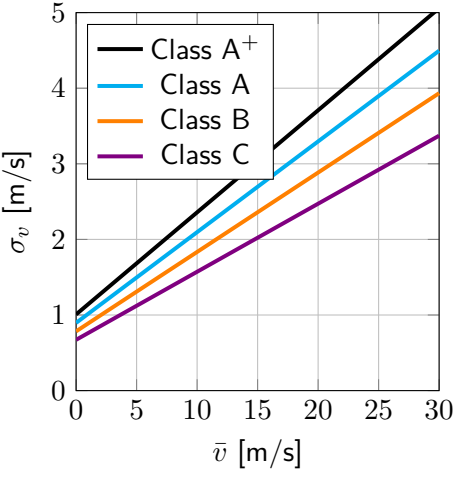
WTG classes acc. to IEC 61400-1 with turbulence classes

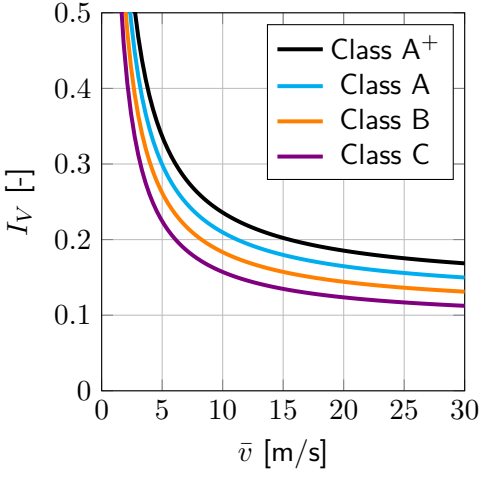
Wind Turbine Class		I	II	III	S
V_{ave}	(m/s)	10	8.5	7.5	Values
V_{ref}	(m/s)	50	42.5	37.5	specified
	Tropical (m/s)	57	57	57	by the
$V_{ref,T}$	(m/s)	57			designer
A+		0.18			
A		0.16			
B		0.14			
C		0.12			

Extract from [IEC 61400-1]

V_{ref}	reference wind speed
$V_{\text{ave}} = 0.2V_{\text{ref}}$	annual mean wind speed at hub height
I_{ref}	expected value of the turbulence intensity at $15\,\text{m/s}$
A^+, A, B, C	turbulence class
σ_v	standard deviation of longitudinal wind component at hub height
v_{hub}	wind speed at hub height

Normal turbulence model (NTM) acc. to IEC 61400-1

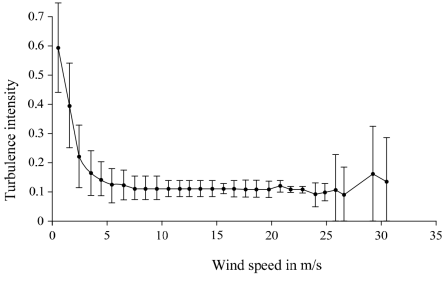




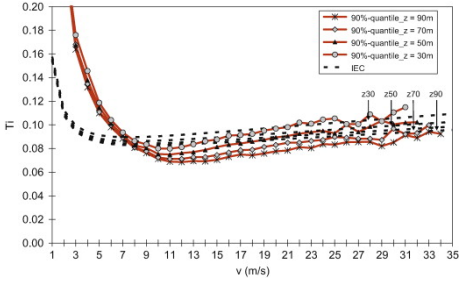
$\sigma_v = I_{\mathrm{ref}}(0.75\bar{v} + b)$ with $b = 5.6\,\mathrm{m/s}$

$I_V = \frac{\sigma_v}{\bar{v}}$

Turbulence intensity from measurements



[1]



[Matthias Türk, Stefan Emeis]

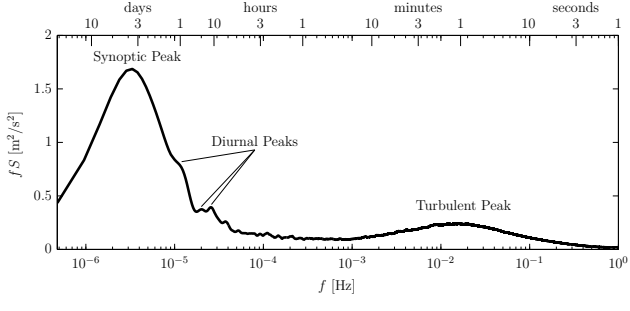
Onshore

- I_V is reduced with increasing wind speed.
- At low wind speeds I_V is strongly dependent on the atmospheric stability.

Offshore

- Wind speed is influencing wave formation and by that the roughness.
- Roughness and wave height are usually increasing with the wind speed.
- Offshore wind turbines are usually simulated with site-specific TI (Class S)

Spectral gap



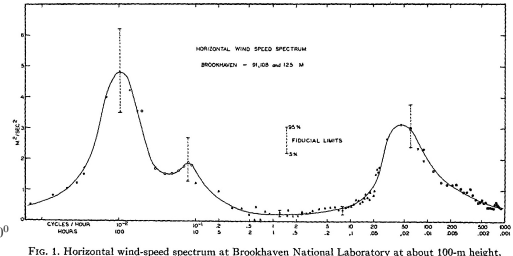
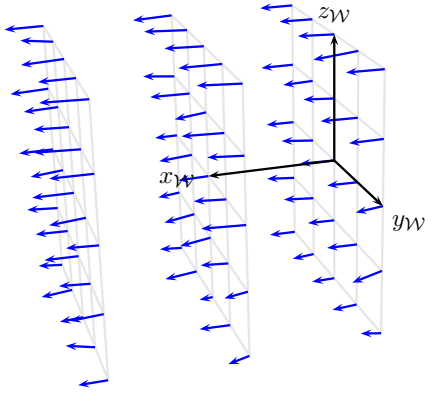


FIG. 1. Horizontal wind-speed spectrum at Brookhaven National Laboratory at about 100-m height. (See table 1 for date and time.)

- Wind spectrum from Bremerhaven measured at 44 m, averaged over 4 months [7]
- The peaks at around 10 and 14 hours are caused by land and sea breezes.
- The wind speed variations with a timescale of 10 minutes or less are known as turbulence.
- The spectrum drops before the turbulence region as in [8].

Introduction to turbulence models



"When I meet God, I am going to ask him two questions: Why relativity? And why turbulence? I really believe he will have an answer for the first."

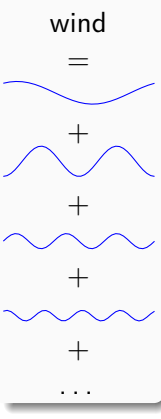
- Werner Heisenberg -

IEC Kaimal spectral model [6]

- Spectra for each grid point i:
  - Longitudinal $S_{ii,u} = \frac{\sigma_u^2 4 L_u/\bar{u}}{(1+6fL_u/\bar{u})^{5/3}}$
  - Lateral $S_{ii,v} = \frac{\sigma_v^2 4 L_v/\bar{u}}{(1+6fL_v/\bar{u})^{5/3}}$
  - Vertical $S_{ii,w} = \frac{\sigma_w^2 4 L_w/\bar{u}}{(1+6fL_w/\bar{u})^{5/3}}$
- Longitudinal coherence between point i and j:

$$\gamma_{ij,u} = \exp\left(-12\sqrt{\left(\frac{f}{\bar{u}}\right)^2 + \left(\frac{0.12}{L_u}\right)^2} r_{ij}\right)$$

Turbulent wind field generation



Wind components without correlation to other points (lateral&vertical IEC Kaimal)

$$v_i(t) = \bar{v} + \sum_{m=1}^{n_\mathrm{f}} a_m \cos(2\pi f_m t + \theta_{i,m})$$

Amplitudes $a_m = \sqrt{2S_{ii,v}(f_m)\Delta f}$ defined by auto-spectrum $S_{ii,v}$ for each grid point i at frequency f_m and discretization Δf. Also single point longitudinal.

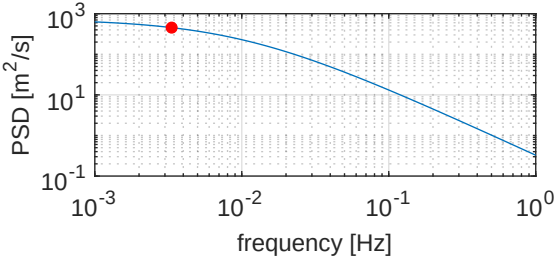
Wind components with correlation to other points (longitudinal IEC Kaimal)

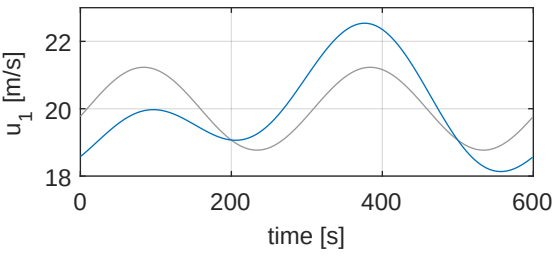
$$u_i(t) = \bar{u} + \sum_{k=1}^{i} \sum_{m=1}^{n_\mathrm{f}} a_{ik,m} \cos(2\pi f_m t + \theta_{k,m})$$

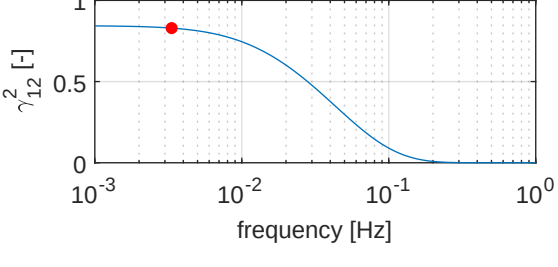
Amplitudes $a_{ik,m}$ defined by auto-spectra and coherence between point i and k.

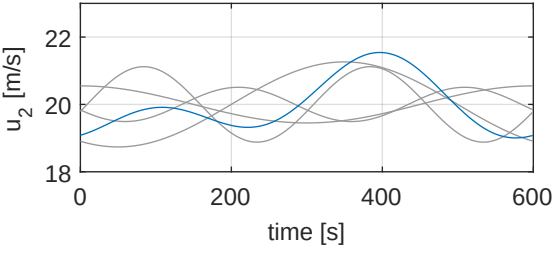
- Wind is "just" a sum of randomly shifted sinusoids with defined amplitudes.
- This method is called the Sandia method or Veers method [9].

Example: longitudinal wind speed in two points

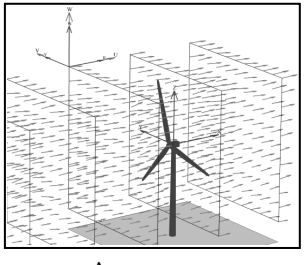


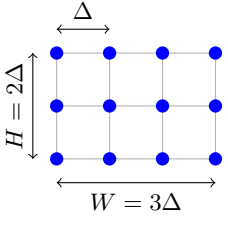






Wind field generation with TurbSim



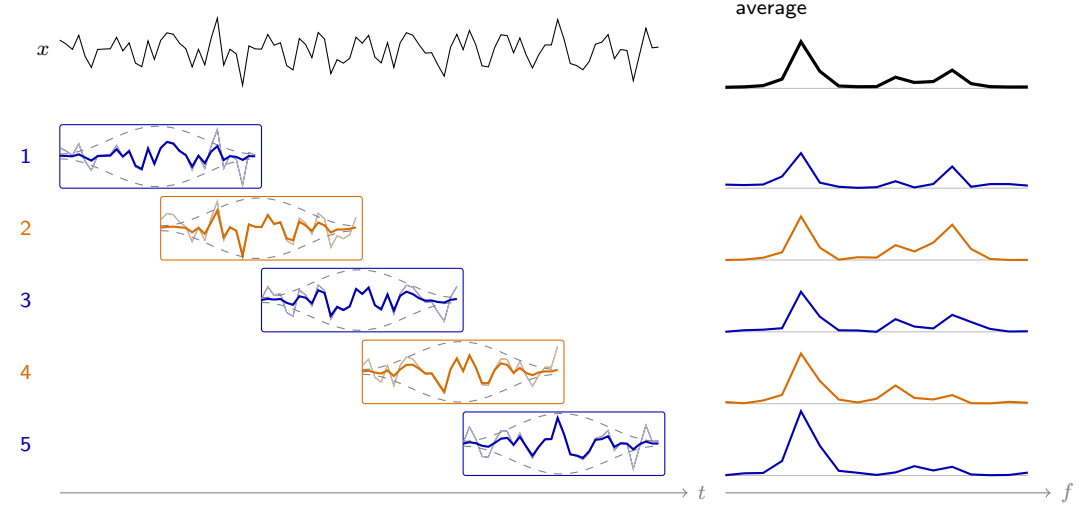


$$n_y = W/\Delta + 1 = 3 + 1 = 4$$
$$n_z = H/\Delta + 1 = 2 + 1 = 3$$

- Wind field generators such as TurbSim [10] usually use seeds to initialize a pseudorandom number generator.
- With this seed, the same wind field can be generated on different PC to generate reproducible results.
- Matlab default is the Mersenne Twister. In TurbSim, usually the RANLUX algorithm is used.
- sowento approach: mean wind speed $\times 100 +$ number from 1 to 6 to generate 6 seeds for each mean wind speed bin.
- If you want to simulate for $600\,\mathrm{s}$ excluding settling time, then use 600 for `AnalysisTime` and `"ALL"` for `UsableTime`. Wind fields are cyclic and "wrap around".
- Avoid the "fencepost error": the number of vertical or horizontal grid points is equal to the grid height or width divided by the resolution plus one.

Power Spectrum Estimation: Welch's Method

5. Average the 5 periodograms.



Spectrum of the rotor-effective wind speed

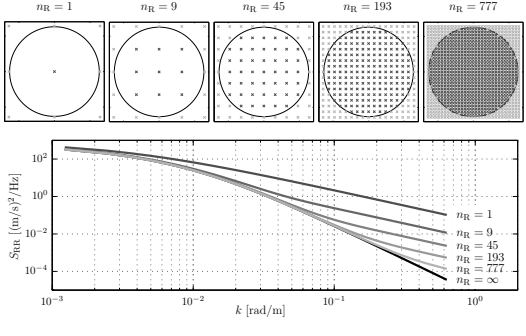
Auto-spectrum of $v_0 = \frac{1}{n}\sum_{i=1}^{n} u_i$

$$S_{\mathrm{RR}} = \mathcal{F}\{v_0\}\mathcal{F}^*\{v_0\}$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\underbrace{\mathcal{F}\{u_i\}\,\mathcal{F}^*\{u_j\}}_{S_{ij,u}}$$

$$= \frac{S_{ii,u}}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\gamma_{ij,u}$$

spectra $S_{ii,u}$ and coherence $\gamma_{ij,u}$ from turbulence model, e.g. Kaimal



[7]

Long-term research challenges

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WIND ENERGY SCIENCE

Long-term research challenges in wind energy – a research agenda by the European Academy of Wind Energy

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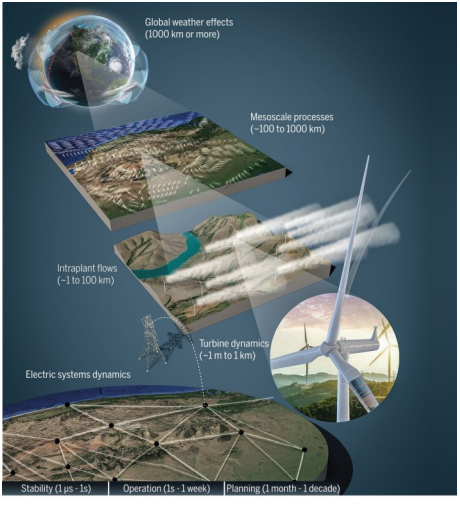
Abstract. The European Academy of Wind Energy (eawe), representing universities and institutes with a significant wind energy programme in 14 countries, has discussed the long-term research challenges in wind energy. In contrast to research agendas addressing short- to medium-term research activities, this eawe document takes a longer-term perspective, addressing the scientific knowledge base that is required to develop wind energy beyond the applications of today and tomorrow. In other words, this long-term research agenda is driven by problems and curiosity, addressing basic research and fundamental knowledge in 11 research areas, ranging from physics and design to environmental and societal aspects. Because of the very nature of this initiative, this document does not intend to be permanent or complete. It shows the vision of the experts of the eawe, but other views may be possible. We sincerely hope that it will spur an even more intensive discussion worldwide within the wind energy community.

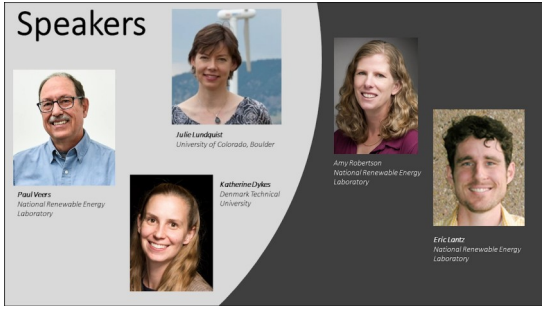
[11]

Covered topics

1. Materials and structures
2. Wind and turbulence
3. Aerodynamics
4. Control and system identification
5. Electricity conversion
6. Reliability and uncertainty modelling
7. Design methods
8. Hydrodynamics, soil characteristics and floating turbines
9. Offshore environmental aspects
10. Wind energy in the electric power system
11. Societal and economic aspects of wind energy

Grand challenges in the science of wind energy 1/2





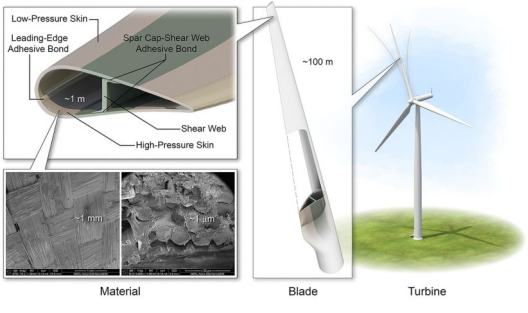
[Video on YouTube]

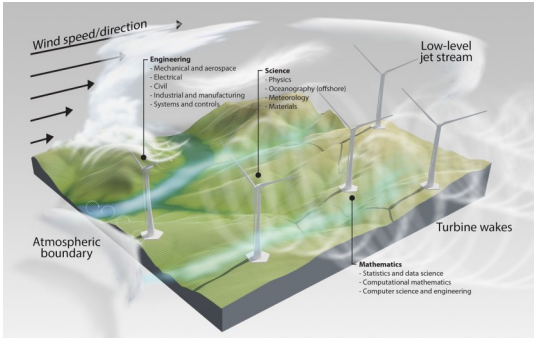
Main challenges

- Mastering the physics of the resource from the atmosphere to the intra-plant flows

[12]

Grand challenges in the science of wind energy 2/2





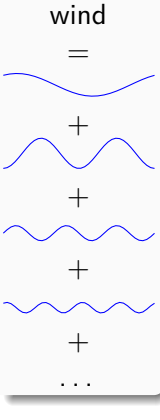
[12]

- Characterizing the structural, aero- and hydrodynamics of the largest rotating structures ever built coupled with access to the most advanced materials at commodity prices
- Controlling wind power plants to orchestrate wind turbine, plant, and grid formation operations for a stable, resilient, reliable and affordable energy system

Conclusion

Main questions

- How is a turbulent wind field generated?



Sum of randomly shifted sinusoids with defined amplitudes

- Lateral, vertical, single point longitudinal IEC Kaimal: amplitudes defined by auto-spectrum, one random phase angle per frequency

$$v_i(t) = \bar{v} + \sum_{m=1}^{n_\mathrm{f}} a_m \cos(2\pi f_m t + \theta_{i,m})$$

- Longitudinal IEC Kaimal: amplitudes defined by auto-spectrum and coherence, phase angles from previously generated points are included

$$u_i(t) = \bar{u} + \sum_{k=1}^{i} \sum_{m=1}^{n_\mathrm{f}} a_{ik,m} \cos(2\pi f_m t + \theta_{k,m})$$

Conclusion

Main questions

- How can we generate rotor-effective wind speed?

Use rotor-effective wind speed spectrum

$$S_{\mathrm{RR}} = \mathcal{F}\{v_0\}\mathcal{F}^*\{v_0\} = \frac{S_{ii,u}}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\gamma_{ij,u}$$

- Generate S_{RR} with spectra $S_{ii,u}$ and coherence $\gamma_{ij,u}$ from turbulence model, e.g. Kaimal
- Use Sandia method to generate a time series

Quick check on learning objectives

After this lecture you should be able to...

- ...describe the origin of wind (global and local effects).
- ...calculate the change of wind speed over height with different models.
- ...generate a time series of longitudinal wind following IEC standard.
- ...name the current research challenges in wind energy.

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Please let me know if you have further questions!

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